Recursive Estimation of ARX Model with Parameters Dependent on High-Dimensional Discrete-Valued Regressors

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The Problem

Motivation: Physical Machine Settings

Materials science experiments involve expensive, time-consuming physical runs.

The Problem: Very low amounts of data relative to the configuration space.

Standard regression models and NNs overfit and/or fail to capture **localized** variances.

Quite common problem of learning with small amounts of data.

Problem Definition: g, z, y

Input Features

Settings Vector `g`: long and categorical

e.g., `[frequency, substrate, operator, ...]`

Regressors `z`: short and numerical

Physical continuous inputs (initial and working pressure).

Output Target

Response `y`: numerical (scalar)

Observed physical outcome of the setting (thickness).

Learning of $p(y \mid z, g, \Theta)$ needs too much data

Approach: Pooled Sub-Models

Instead of one complex model, we build a **Pool of 'Experts'**:

Sub-Model Unit

For each category `j`, in each entry `i` of `g`: `g_ij`.

Simple 1D Linear ARX Model.

Initiated with **non-informative priors**.

Adaptive Learning

- Updates via **recursive Bayesian estimation**.
- Employs **Optimal Forgetting**.
- Adapts to the system automatically.
- Calculates sub-model's sufficient statistics.

The Combination Logic

Single predictive distribution is a weighted geometric mean of individual predictors:

$$p(y \mid z, g, \text{data}) \propto \prod_{\text{i,j}} p(y \mid z, g_{ij}, \text{data})^{\phi_{\text{ij}}}$$

Combining sub-models into a unified forecast by **minimizing relative-entropy**:

$$F(\phi) = \text{KLD}(\phi \parallel \phi_0) - \int \prod_{\text{i,j}} p(y \mid z, g_{ij}, \text{data})^{\phi_{\text{ij}}} dy$$

The **optimal weights** ϕ minimize F(ϕ)

The Learning Goal: Recommendation of settings

Inverse Optimization

Search for the **optimal g**.

Given a target response `y*`.

Exhaustive search over `g` entries maximizing p(y* | z, g, data).

Recommends settings to achieve the physical goal.

Validation: Leave-One-Out (LOO)

Methodology

Hold out one observation (`y, z, g`).

Train the combined model on all remaining N-1 points.

Predict the held-out point using found model.

Repeat for all N observations.

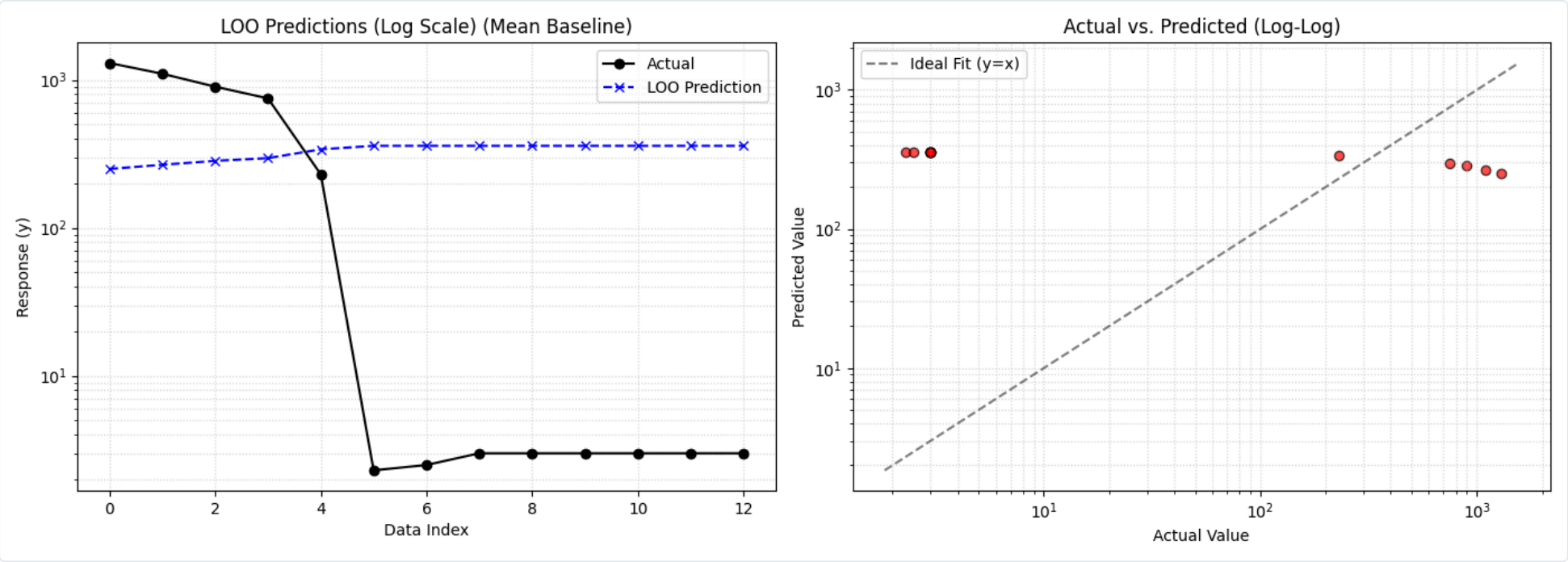
Benchmarked against two standard references:

Mean Baseline: Average of N-1 observations.

Simple OLS: Ordinary Least Squares on continuous `z` only.

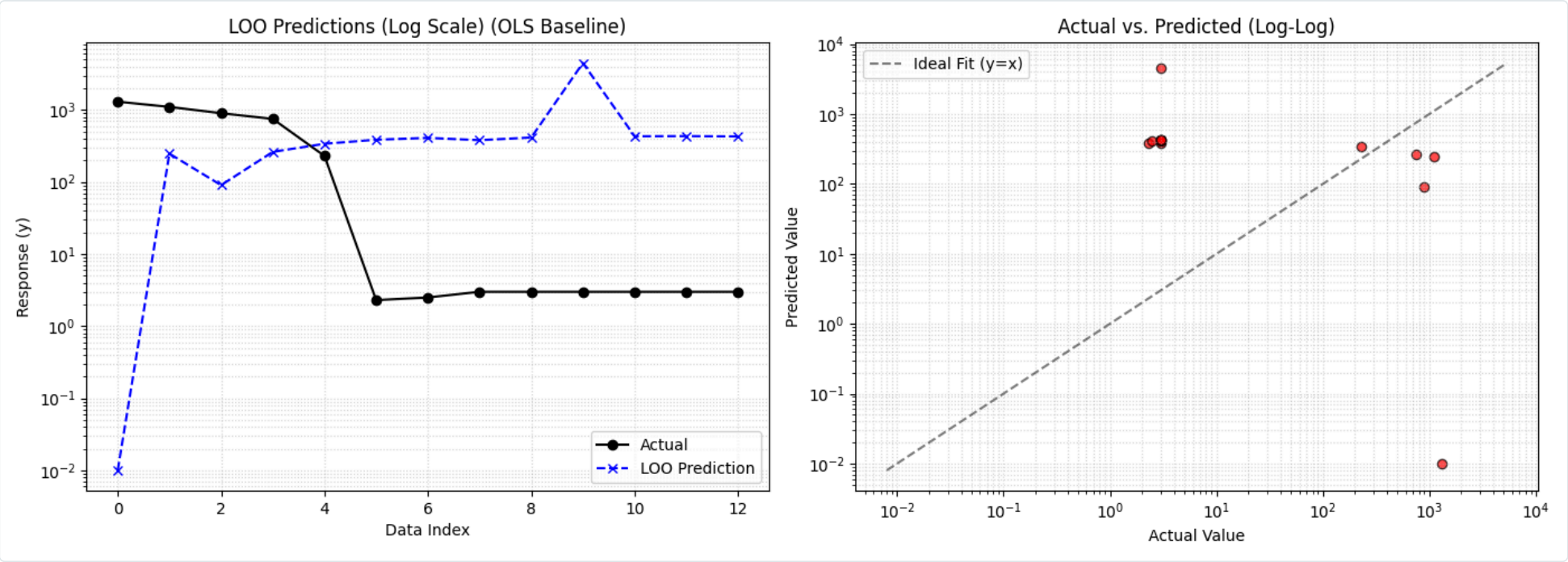
Validation: Baseline Benchmarks (Mean)

Mean Baseline



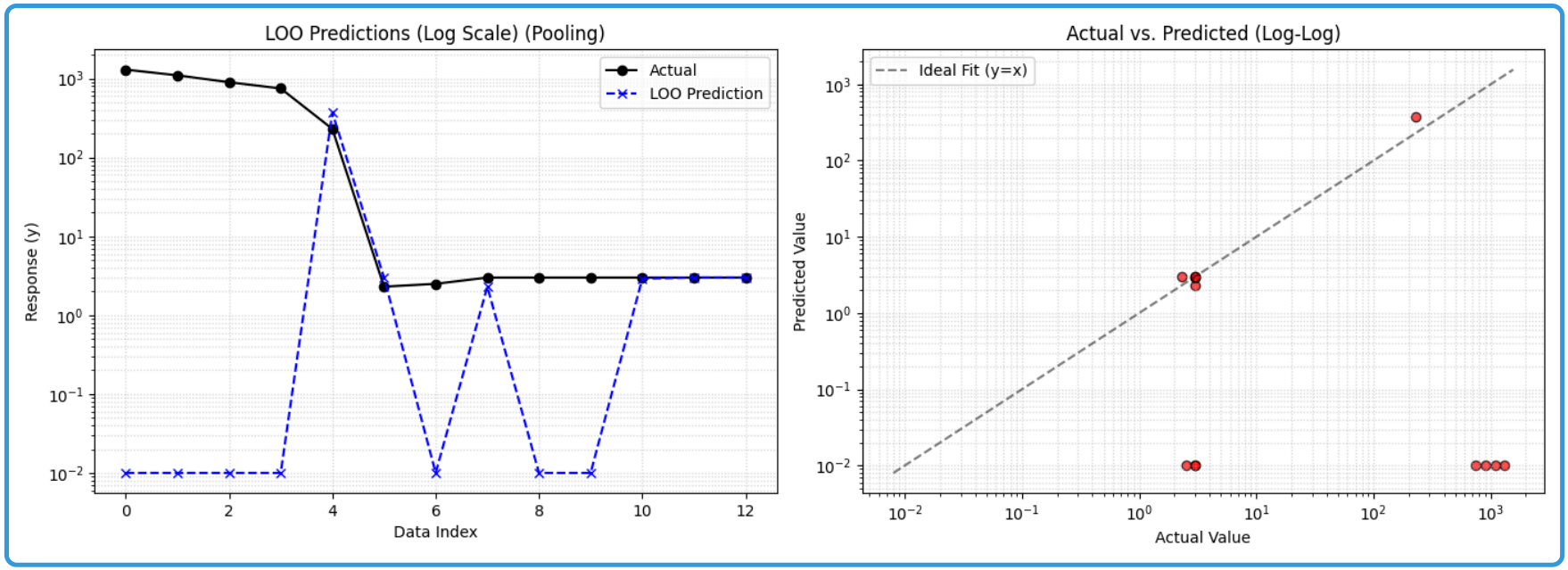
Validation: Baseline Benchmarks (OLS)

Simple OLS



Validation: Pooled Model

Proposed Pooling Approach



Future Work & Conclusions

Work in Progress: Refining the recursive estimation of the ARX parameters and testing robustness on larger categorical dimensions.

Hierarchical Priors: Worth exploring to share information across categories with physical similarities, further mitigating data scarcity.

A Universal Challenge: This is a hard and common problem across multiple high-stakes industries where data collection is expensive:

- **Semiconductors:** Photolithography settings and wafer variability.
- **Aerospace:** Turbine blade alloy composition and additive manufacturing.
- **Pharmaceuticals:** Batch process optimization and bioreactor control.
- **Precision Engineering:** CNC machining tolerances with variable tool wear.

Thank you

Questions?

Citations

Kárný Miroslav: Optimised conjugate prior for model structure estimation in the exponential family, Expert Systems With Applications, DOI: 10.1016/j.eswa.2025.127716 [2025]

Kárný Miroslav: Optimized geometric pooling of probabilities for information fusion and forgetting, Automatica 177 DOI: 10.1016/j.automatica.2025.112337 [2025]

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